

Effects of cable attachment position and pulling direction on gait biomechanics during walker-supported walking

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Abstract

Background. Cable-driven gait-assistive devices have so far been developed as standalone soft exosuits anchored at the hip or ankle. Wheeled walkers, the most widely prescribed mobility aid for older adults and for individuals after stroke, primarily provide passive mechanical support. Integrating cable-driven assistance into the walker frame combines the stability of the walker with the active swing-phase support of an exosuit, but the appropriate cable attachment site on the leg and the appropriate pulling direction relative to the walker have not been characterized.

Methods. Twelve healthy adult men walked on an instrumented treadmill at 1.0 m/s while supported by a custom walker that maintained 20 % body-weight unloading through bilateral instrumented handrails with visual biofeedback. A Bowden cable, mounted on the walker frame and actuated by a brushless motor, applied a half-sine force pulse (peak 50 N, 55–85 % of the gait cycle, peak at 70 %) to the participant’s dominant-side shank during each stride. Six conditions were tested in a counterbalanced order: three cable attachment positions on the shank (proximal, mid-shaft, distal — at 25, 50, and 75 % of shank length) crossed with two pulling directions (0° horizontal and 30° oblique). Surface electromyography of five lower-limb muscles, full-body kinematics, ground reaction force, cable force, and subjective comfort ratings were recorded. Two-way repeated-measures ANOVA tested the main effects and interaction of position and direction.

Results. *Results will be reported following completion of data collection (n=12 healthy adult men).*

Conclusions. This study characterizes the position- and direction-dependent biomechanical effects of cable assistance under walker support and is intended to inform parameter selection for subsequent walker-mounted assistance studies.

Keywords: walker; cable-driven exosuit; gait assistance; body weight support; biomechanics; electromyography

1 Background

Wheeled walkers are the most widely prescribed mobility aid for older adults and for individuals recovering from stroke or Parkinson’s disease. They reliably provide a stable base of support and reduce the risk of falls, but their assistance is primarily passive: the walker frame does not contribute energy to the user’s swing-phase limb advancement. Long-term walker users develop

a characteristic gait pattern of reduced velocity, shortened stride length, prolonged stance time, and reduced swing time [1, 2], and this pattern can itself contribute to deconditioning. The walker frame, however, is a structurally rigid platform positioned in front of the user; its use as an anchor for active swing-phase assistance has received limited systematic evaluation.

Cable-driven actuation has emerged over the past decade as a widely used approach to soft, wearable gait assistance [2, 3]. In hip-targeted soft exosuits, a cable anchored to a waist belt and routed to a thigh brace produces hip-flexion or hip-extension assistance; the timing of this assistance has been identified as a critical determinant of metabolic and biomechanical benefit [4, 5]. In ankle-targeted devices, cables or elastic mechanisms acting across the ankle can reduce the metabolic cost of walking by 7–21 % in healthy adults [6, 7] and have demonstrated translational benefit for stroke survivors with assistance corresponding to roughly 12 % of the biological joint torque [8]. Multi-joint configurations spanning the hip, knee, and ankle have also been characterized [9–11].

In all of these systems, the cable’s proximal anchor sits on the user’s body, and the cable’s reaction force is transmitted through another segment of the user. A walker-mounted cable, in contrast, anchors its proximal end to the walker frame; the cable’s reaction force is borne by the walker rather than by the user. This changes both the geometry of the cable’s lever arm and the available range of pulling directions, since the anchor’s height on the walker frame can be modified to alter the cable’s angle of action at the shank attachment site. We are not aware of prior studies that systematically characterized the effects of cable attachment position on the shank and the cable pulling direction in this walker-integrated configuration. Recent work on conventional soft exosuits has reported that anchor-point selection can have a non-monotonic effect on force capability and human–suit stiffness [12], and broader work on cable-driven exoskeletons has indicated that subject-specific attachment-point optimization may be important for matching natural muscle coordination [13].

The aim of the present study was to characterize the effects of three shank attachment positions and two cable pulling directions on lower-limb muscle activity, joint kinematics, ground reaction forces, and subjective comfort during walker-supported treadmill walking with 20 % body-weight unloading. The cable force was applied as a half-sine pulse spanning 55–85 % of the gait cycle, with the onset placed during pre-swing so that the cable engages the limb during the rising phase of natural shank velocity rather than initiating motion from rest at toe-off. We expected the cable’s effect on muscle activity and kinematics to scale with attachment position because the cable’s moment arm to the hip and knee changes monotonically along the proximal–distal axis of the shank, and to differ between pulling directions because a more oblique cable redistributes a greater fraction of its force into a vertical lift component. The results are intended to inform the parameter selection of future studies of walker-mounted cable-driven gait assistance.

2 Methods

2.1 Participants

Twelve healthy adult men (age 20–45 years, body mass 60–90 kg, height 165–190 cm) were recruited. All participants were able to walk on a treadmill at 1.0 m/s for at least five continuous minutes without an assistive device. Participants reported no current or recent (within six months) lower-limb injury or surgery, no diagnosed neurological or cardiovascular condition, and no prior exposure to a cable-driven exoskeleton or exosuit within the preceding twelve months. The cohort was restricted to male participants to reduce inter-subject anthropometric variance, which kept the cable’s lever-arm geometry comparable across participants. This restriction is considered when interpreting the generalizability of the sample. The study protocol was reviewed and approved by the Institutional Review Board of Chung-Ang University, and all participants provided written informed consent prior to participation.

Figure 1 placeholder. Schematic of the integrated walker-mounted cable-driven assistive system: instrumented treadmill (Bertec TM-09-P); custom walker frame with bilateral handrail load cells and visual biofeedback display for the 20 % body-weight unloading target; Bowden cable assist module driven by the AK60-6 motor with the cable terminating at one of three shank attachment positions (HIGH/MID/LOW) and exiting the walker frame through anchor points selectable for two pulling directions ($0^\circ/30^\circ$); Qualisys motion-capture cameras; Delsys Trigno wireless EMG sensors on the dominant lower limb; foot-mounted EBIMU24GV6 inertial measurement unit. To be replaced with the rendered illustration prior to submission.

Figure 1: Schematic of the integrated walker-mounted cable-driven assistive system used in the present study.

The sample size was determined a priori using G*Power 3.1 [14] for a within-factors repeated-measures ANOVA. Setting $\alpha = 0.05$, statistical power $1 - \beta = 0.80$, an assumed correlation among repeated measures of 0.5, and a moderate-to-large effect size $f = 0.40$, the required sample size was 12 participants. The chosen effect size is consistent with within-subject effects reported in comparable cable-driven exoskeleton studies in healthy adults, in which sample sizes of 5–10 detected significant condition-level differences in muscle activity and joint kinematics [3, 7, 9]. Up to 15 participants were recruited to accommodate dropouts.

2.2 Apparatus

A schematic of the integrated apparatus is shown in Figure 1.

Walking was performed on an instrumented split-belt treadmill (TM-09-P, Bertec, Columbus, OH, USA) at a constant belt speed of 1.0 m/s, with bilateral ground reaction forces sampled at 2000 Hz. Three-dimensional kinematics were recorded with an optoelectronic motion-capture system (Qualisys, Göteborg, Sweden) at 200 Hz using a marker set that covered both the lower limbs and the trunk, allowing trunk lean and pelvis dynamics — both of which are characteristically altered during walker-supported walking — to be tracked together with lower-limb joint angles. Joint kinematics and inverse-dynamic estimates of joint moments and powers were computed offline using musculoskeletal-modelling software (Visual3D, HAS-Motion, Kingston, ON, Canada).

Surface electromyography was recorded with a wireless system (Trigno Wireless, Delsys, Natick, MA, USA) from five muscles of the dominant lower limb. Electrodes were placed at SENIAM-recommended landmarks [15]: gluteus maximus (GMax) at 50 % of the line from the sacral vertebra to the greater trochanter; rectus femoris (RF) at 50 % of the line from the anterior superior iliac spine to the superior pole of the patella; biceps femoris long head (BF) at 50 % of the line from the ischial tuberosity to the lateral epicondyle of the tibia; tibialis anterior (TA) at 1/3 of the line from the tip of the fibular head to the medial malleolus; and medial gastrocnemius (MG) over the most prominent bulge of the medial gastrocnemius head. The five-muscle set targets the principal hip extensor (GMax), the biarticular hip-flexor / knee-extensor (RF), the biarticular hip-extensor / knee-flexor (BF), the ankle dorsiflexor (TA), and the principal ankle plantarflexor (MG). Skin preparation followed standard practice (shaving when needed and cleaning with isopropyl alcohol), and the placement was verified during muscle-specific manual

resistance tests prior to each session. Maximum voluntary contractions (MVC) were collected for each channel using muscle-specific isometric postures and were used to amplitude-normalize the EMG envelopes. EMG signals were logged at 200 Hz on the host computer.

The walker (Figure 1) was a custom rigid frame placed in front of the treadmill. Each handrail of the walker was instrumented with a single-axis load cell that measured the vertical force transmitted through the user’s hand. The summed handrail force, expressed as a fraction of the participant’s body weight, was displayed as a real-time bar graph at the participant’s eye level, and participants were instructed to maintain the bar within a target band centred at 20 % of body weight ($\pm 2\%$). Trials in which the time-averaged handrail-supported fraction deviated from the 20 % target by more than the tolerance were flagged for repetition. The 20 % target was selected to lie well within the range over which body-weight unloading does not appreciably alter spatio-temporal and kinematic gait parameters in healthy adults [1, 16].

The cable assist module was mounted on the front frame of the walker. The cable was actuated by a brushless DC motor (AK60-6, CubeMars, China) operating in current-control mode through a small-radius pulley, and cable tension was measured at the distal end by an in-line single-axis load cell (LSB205, Futek, Irvine, CA, USA). The cable exited the walker frame through anchor points whose vertical position could be adjusted to produce two pulling directions: 0° (horizontal forward) and 30° (oblique up-forward), defined relative to the horizontal at the cable attachment site in the neutral standing pose. The distal end of the cable terminated in a soft thermoplastic-elastomer sleeve secured to the participant’s dominant-side shank with hook-and-loop straps. Three sleeves of identical compliance were used for the three cable attachment positions, defined as 25, 50, and 75 % of shank length (HIGH, MID, LOW) measured from the medial knee joint line to the medial malleolus and corresponding to the proximal tibia just distal to the tibial tuberosity, the mid-tibial shaft, and the distal tibia just proximal to the malleoli. The motor controller enforced a peak cable-tension safety cutoff at 70 N: if the load-cell reading exceeded this threshold the commanded current was set to zero and the cable was allowed to pay out.

A foot-mounted wireless inertial measurement unit (EBIMU24GV6, E2BOX, Republic of Korea) on the dorsum of the dominant-side shoe provided a continuous, stride-by-stride estimate of gait-cycle phase. Heel-strike events were detected from the vertical acceleration peak, and the elapsed fraction of the previous stride duration was used to interpolate the current percentage of the gait cycle. The cable force envelope was gated to its assigned 55–85 % gait-cycle window using this phase estimate. All measurement streams (motion capture, ground reaction force, EMG, cable load cell, foot IMU, and walker handrail load cells) were time-synchronized through a common analog trigger and brought to a common time-base offline by linear interpolation prior to analysis.

2.3 Experimental conditions

Each participant completed seven walking conditions in a single session. The first condition (WBW) was walker-supported walking with 20 % body-weight unloading and the cable system in passive (zero-tension) mode; this condition served as the within-subject reference. The remaining six conditions (WBAW) corresponded to the factorial combination of three shank attachment positions (HIGH, MID, LOW) and two pulling directions (0° , 30°). The order of the seven conditions was counterbalanced across participants using a balanced Latin square. Comparisons against unsupported normal walking were not pursued in this study, because the kinematic and electromyographic effects of body-weight unloading on healthy gait have been examined in prior work [1, 16]; the present study focuses instead on the additional effects of cable assistance over and above the body-weight-supported baseline.

Within each WBAW condition, the cable tension followed a half-sine envelope with peak amplitude 50 N. The envelope was gated to 55–85 % of the gait cycle on a stride-by-stride basis, with the peak occurring at the geometric centre of the window (70 % GC). The 50 N peak

corresponds to approximately 7% of body weight for a 70 kg participant; integrated over the half-sine window at the typical cable-attachment velocity of mid-swing this corresponds to an average positive mechanical power of approximately 0.55 W/kg per leg, which is in the range explored in prior cable-driven exo studies in healthy adults [7, 9] and well below the shank comfort limit for axial pulling forces reported by Yandell et al. [17]. The half-sine shape was chosen for its smooth onset and release, which avoids impulsive force changes at the boundaries of the actuation window.

The 55–85% timing was chosen to align the cable’s force trajectory with the natural acceleration profile of the swing limb. The 55% GC onset falls in pre-swing, after contralateral heel strike has transferred most of the body weight to the contralateral limb and as the ipsilateral hip flexors begin to activate [18]; at this instant the shank’s forward angular velocity is small but rising, so the cable was intended to build tension while the limb is already moving forward, rather than having to initiate motion from rest at toe-off. The 70% peak was placed at the approximate peak of the shank’s angular velocity in mid-swing, with the intention of placing the cable’s instantaneous mechanical power transfer in the high-velocity portion of the swing. The 85% release was set to precede the natural reversal of the hip moment from net flexor to net extensor in late swing, which has been reported to occur at approximately 86% GC [19]; this was intended to keep the cable’s mechanical work on the limb predominantly in the positive (concentric) regime and to avoid interference with the natural eccentric deceleration of the limb at the end of swing. The same envelope shape, peak amplitude, and gait-cycle window were applied identically across all six WBAW conditions, so that attachment position and pulling direction were the only experimentally manipulated factors. Implementation details of the cable controller — including pre-onset slack management and feedforward components — are reported in the Supplementary Material.

2.4 Protocol

Each session lasted approximately two hours per participant. After consent and screening, body mass, standing height, and leg-segment lengths required for marker placement and cable attachment-site definition were measured. Reflective markers and EMG electrodes were placed, and MVC trials were collected for each EMG channel. Participants then walked on the treadmill for six minutes at 1.0 m/s without the walker as familiarization, which is the minimum duration reported to stabilize the spatio-temporal and kinematic gait parameters in healthy adults [20]. The seven experimental conditions were presented in a counterbalanced order; each trial consisted of 90 s of continuous walking during which all measurement streams were recorded, followed by a rest of at least 90 s during which the participant completed a Borg rating of perceived exertion (CR-6–20), a comfort visual analog scale (0–100), and a naturalness visual analog scale (0–100). The cable attachment position and walker anchor height were reconfigured between trials as required by the upcoming condition. The first ten strides of each trial were excluded from analysis to allow stabilization, and the subsequent thirty consecutive strides were used for outcome computation.

2.5 Outcome measures

Outcomes were organized into a primary–secondary–exploratory hierarchy that was specified prior to data analysis.

Primary outcomes. Two outcomes were designated as primary: (i) swing-phase integrated EMG of the rectus femoris (RF iEMG, %MVC·%GC), selected because the RF is the principal biarticular muscle whose action — hip flexion combined with knee extension during pre-swing and early swing — most closely overlaps the cable’s expected action; and (ii) peak hip flexion angle during swing, selected as a single, kinematically interpretable summary of the cable’s

effect on swing-phase limb advancement. Hypothesis tests for these two primary outcomes were considered confirmatory.

Secondary outcomes. Secondary outcomes included swing-phase iEMG of the remaining four muscles (GMax, BF, TA, MG); peak knee flexion angle, swing range of motion at the hip and knee, swing duration, stride duration, step length, and minimum foot clearance; and the propulsive and braking impulses derived from the anteroposterior ground reaction force. These outcomes were used to characterize the broader biomechanical response and to inform the interpretation of the primary outcomes.

Exploratory outcomes. Exploratory outcomes included sagittal trunk lean (a kinematic indicator of walker-related compensation), the three subjective ratings (Borg RPE, comfort VAS, naturalness VAS), and cable-side measurements (mean and standard deviation of the measured peak cable force per stride and the time of the peak relative to the IMU-estimated stride start; the latter was used to verify force-tracking fidelity and to compute the average positive mechanical power delivered per stride).

EMG processing. The Trigno system performed onboard band-pass filtering and rectification of the raw EMG before downsampling to the 200 Hz host-side log rate. The 200 Hz logged signals were further smoothed with a 4th-order zero-phase-lag low-pass Butterworth filter (10 Hz cutoff) to obtain the linear envelope. Stride epochs were defined from heel strike to ipsilateral heel strike using the foot IMU and verified by the vertical ground reaction force. Each stride's envelope was time-normalized to 101 points spanning 0–100 % GC and amplitude-normalized to the channel-specific MVC peak. The integrated EMG (iEMG) over the swing phase (defined here as 60–100 % GC) was computed for each stride and averaged across the thirty analyzed strides per condition.

2.6 Statistical analysis

Statistical inference was confined to the two pre-specified primary outcomes (RF iEMG and peak hip flexion angle); the secondary outcomes were analyzed with the same procedure but with a Bonferroni adjustment applied across the secondary family to control the family-wise error rate, and the exploratory outcomes were summarized descriptively without inferential claims.

Each primary and secondary outcome was analyzed with a two-way repeated-measures ANOVA across the six WBAW conditions with within-subject factors of cable attachment position (HIGH, MID, LOW) and pulling direction (0°, 30°). Where Mauchly's test indicated a violation of sphericity, Greenhouse–Geisser correction was applied. Effect sizes were reported as partial eta-squared (η_p^2) for ANOVA effects and Cohen's d for pairwise contrasts. Significant main effects or interactions were followed up by pairwise comparisons with Bonferroni correction. The WBW reference condition was compared with the WBAW grand mean and with each WBAW condition using paired-sample t -tests with Bonferroni correction. Statistical significance was set at $\alpha = 0.05$ for the primary outcomes and at the Bonferroni-adjusted level for the secondary family.

Trial-validity criteria, the outlier rule (within-subject values exceeding ± 3 SD of the participant's own condition mean were inspected; values associated with identifiable measurement-system errors were excluded, while physiological outliers were retained), and the analysis pipeline were specified prior to data collection and registered on the Open Science Framework. All analyses were performed in MATLAB (R2025b, MathWorks, Natick, MA, USA) or Python (NumPy, SciPy, statsmodels), as appropriate to each analysis stage. The analysis code and de-identified data will be deposited in a public repository (GitHub with a Zenodo digital object identifier) upon acceptance of the manuscript.

3 Results

Pending data collection.

4 Discussion

Principal findings

Pending data collection. Will be filled with the position $\times$ direction effect on each primary outcome, the candidate condition that best balances reduced muscular effort with preserved gait quality, and the corresponding effect sizes.

Mechanism of action

The cable applied at the shank produces a hip-flexion moment whose magnitude scales with the perpendicular distance from the hip joint center to the cable’s line of action — that is, with the vertical position of the attachment site below the hip. The same cable also produces a knee moment, whose magnitude scales with the attachment-to-knee distance and whose direction is predominantly knee-extension because the cable’s line of action passes anterior to the knee in the swing posture. As the attachment moves distally along the shank, both moment arms increase, and the ratio of knee to hip torque shifts in favor of the knee. The 0° horizontal direction maximizes the horizontal force component and thus the knee leverage, whereas the 30° oblique direction redistributes part of the force into a vertical lift component that augments hip elevation while reducing the effective knee leverage.

This geometry suggests a specific functional analogy. The combination of a hip-flexion moment and a knee-extension moment is the characteristic action of the rectus femoris during pre-swing and early swing, and the cable’s action may provide an external mechanical analogue of part of this biarticular function. The 55–85 % GC actuation window also overlaps with the period over which the rectus femoris is naturally active in healthy gait, so the cable’s force is in phase with the muscle’s natural recruitment rather than out of phase with it. Prior studies suggest that swing-phase assistance produces larger biomechanical effects when its onset and offset are matched to the user’s natural kinematic and muscular cycle [4, 7].

Comparison with prior work

The peak cable forces used in the present study (50 N) and the corresponding average positive mechanical power (approximately 0.55 W/kg per leg) lie within the range explored in prior cable-driven exo studies in healthy adults. The peak shank tension is approximately 7 % of the body-weight-normalized comfort limit reported by Yandell et al. [17] for axial pulling at the shank, remaining below the reported shank comfort threshold. The 12 % biological-torque level used by Awad et al. [8] to restore paretic-ankle dorsiflexion in stroke survivors and the 24 % biological-torque optimum reported by Kim et al. [5] for hip-flexion exosuits in healthy adults are within the same order of magnitude as the present device’s setpoint. The placement of the actuation window at 55–85 % GC, and the placement of the peak in mid-swing, are consistent with timing strategies reported to produce favorable metabolic and biomechanical outcomes in prior hip- and ankle-targeted devices [4, 7].

Limitations

The cohort was restricted to healthy adult men. Although this restriction reduced inter-subject anthropometric variance and made the cable’s lever-arm geometry more comparable across participants, it limits the immediate generalizability of the findings to other populations. Sex-related differences in pelvis kinematics, lower-limb segment proportions, and muscle activation

patterns during gait are documented in the literature and may modulate the position- and direction-dependent effects characterized here. The findings should therefore be interpreted as specific to the male cohort tested, and replication in female participants and in older adults representative of typical walker users is needed before generalizing the candidate condition to those populations.

The study was performed at a single belt speed and with a single peak cable force; a comprehensive dose response across forces and a speed response across walking velocities were beyond the scope of this initial characterization and are identified as priorities for follow-up work. Although a wider range of pulling directions — including a more oblique angle compatible with the LOW attachment geometry of the walker frame — was considered during pilot testing and could be incorporated in subsequent studies, the present design was restricted to two directions to keep the within-subject burden manageable. The study was also conducted on a treadmill at a single fixed speed, and the documented differences between treadmill and overground walking biomechanics may limit the direct overground generalization of the findings.

Future work

The position- and direction-dependent effects characterized here may inform parameter selection for subsequent walker-mounted cable-driven assistance studies, and the load-cell-based visual biofeedback paradigm could be adapted into a closed-loop unloading controller in subsequent system iterations. The parameter map obtained in healthy adults may guide subsequent evaluation in clinical cohorts. A wider exploration of the position-direction design space — including a third pulling direction at LOW attachment to take advantage of the steeper cable angle that the walker frame geometrically permits — could be examined once the present preliminary characterization has narrowed the parameter space.

5 Conclusions

Pending data collection. Will summarize the candidate optimal attachment position and pulling direction for the walker-mounted cable-driven assist, the corresponding reductions in muscle activity and changes in gait kinematics, and the implications for the design of the next-generation device.

Declarations

Ethics approval and consent to participate. The study protocol was reviewed and approved by the Institutional Review Board of Chung-Ang University. All participants provided written informed consent prior to participation.

Consent for publication. Not applicable.

Availability of data and materials. The de-identified datasets and the analysis code generated during the present study will be deposited in a public repository upon publication; the DOI will be inserted at the proof stage.

Competing interests. The authors declare that they have no competing interests.

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